

Evolutionary Algorithm In HPC Model For Protein Structure Prediction

Shih-Chieh Su and Jyh-Jong Tsay

Department of Computer Science and Information Engineering

National Chung Cheng University, Chiayi County, Taiwan

{ ssc95p, tsay }@ cs.ccu.edu.tw

Abstract

The Proteins play fundamental and crucial roles in nearly all biological processes, such as, enzymatic catalysis, signaling transduction, embryonic development, DNA and RNA synthesis and et al. It has been a long standing goal in molecular biology to predict the tertiary structure of a protein from its primary amino acid sequence. The renowned hydrophobic-Polar (HP) model provides a computation efficient way to generate a reference structure. However, the HP model is can be improve because it only takes the hydrophobic and the polar into account. This paper proposes a new hydrophobic-polar-cysteine (HPC) model which is integrated with the evolutionary algorithm to acquire a more accurate result. More specifically, the disulfide bond is mixed into the fitness function. Additionally, we propose lattice rotation for crossover and K-site move for mutation, which form the key components of our evolutionary algorithms. Experiment results show that the proposed approach is able to find minimum-energy conformations. In addition, they show that in the presence of disulfide bonds the HPC model outperforms the HP model in terms of RMSD measure. It suggests that the proposed approach can improve similarity between the predicted result and the real protein structure under the condition that proteins contain disulfide bonds on the ab initio in protein structures.

1 Introduction

Proteins are essential biological molecules playing vital roles in nearly all biological processes. Therefore the prediction of a protein's tertiary structure based on its primary amino acid sequence has long been the most important and challenging subject in biochemistry, molecular biology and biophysics.

The non-covalent interactions are the driving forces behind the formation of 3D tertiary structure of a protein which include hydrophobic interaction, hydrogen and electrostatic bonds. Among the various interactions, the main driving force in the protein 3D tertiary structure is a tight

hydrophobic core formed by the hydrophobic amino acids of the protein which stabilizes its structure [1]. As a result, Lau and Dill in 1989 first introduced a simple hydrophobic-Polar model (HP model) [2] based on the hydrophobic interaction between amino acids which has greatly reduced the complicity in protein structure predication. HP model though simple, has been shown to be NP-complete for the protein structure prediction problem even in the simple 2D square lattice [3] and 3D cubic lattice [4].

Another important and stabilizing force in protein folding is the internal disulfide bonds [5]. A disulfide bond is covalent as formed by two cysteine residues and contributes significantly to the formation of a stable 3D structure. Consequently to be able to predict correctly the disulfide bonds is an important topic in the computational prediction of the protein structure. To this date there have been groups who have furthered their research by investigating the influence of adding disulfide bonds into the HP model such as, for example, the hydrophobic-polar-cysteine (HPC) model proposed by [6-7]. Khodabakhshi et al. from the point view of the design principle discussed further the application HPC model on 2D square lattice [6] and 3D hexagonal prism lattice [7] to increase the stability of the protein structure and also proved that the insertion of disulfide bonds helped to form a stable protein structure of a designed peptide.

In our study the addition of important disulfide bonds in our HPC model was developed to study the optimization problem in protein folding or protein structure prediction. The experimental results show that HPC model with the presence of disulfide bonds performed much better on RMSD measure than HP model which ignores the presence of disulfide bonds. This suggests that 3D FCC HPC model can predict a protein structure closer to its natural folding state in HP model.

2 Preliminaries

In this section, the background of this study is discussed, including HP model fitness function, HPC model fitness function and 3D FCC lattice

model.

2.1. FCC Lattice Model

Raghunathan and Jernigan [8] in 1997 introduced a 3D FCC lattice model. This lattice can readily be used in conformation simulations to reduce the conformational space. Consequently, the 3D FCC lattice can be described by a diagram as in Figure 1. In the 3D FCC lattice model, each lattice point has 12 neighbors representative of {FL (+1,+1,0); FR(+1,-1,0); FU(-1,+1,0); FD(-1,-1,0); BL(+1,0,+1); BR(+1,0,-1); BU(-1,0,+1); BD(-1,0,-1); LU(+0,+1,+1); LD(+0,+1,-1); RU(+0,-1,+1); RD(+0,-1,-1)}. The numbers 1-12 represents 12 positions in lattices.

In our study, the symbols have absolute positions. The symbols FL;FR;FU;FD;BL;BR;BU;BD;LU;LD;RU; and RD are used to denote the fold directions: FL is for front-left, FR is for front-right, FU is for front-up, FD is for front-down, BL is for back-left, BR is for back-right, BU is for back-up, BD is for back-down, LU is for left-up, LD is for left-down, RU is for right-up and RD is right-down in the encoding scheme, respectively.

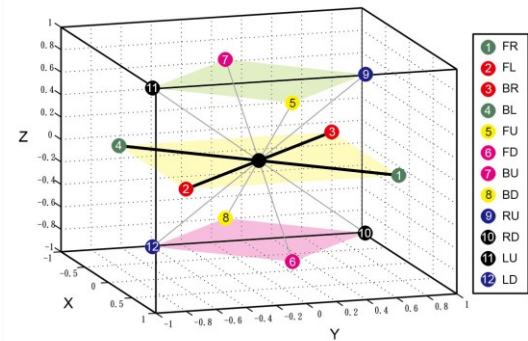


Figure 1. The FCC lattice model. Each lattice point has 12 neighbors representative of {1-FL;2-FR;3-FU;4-FD;5-BL;6-BR;7-BU;8-BD;9-LU;10-LD;11-RU;12-RD}.

2.2. HP Model

The fitness function is at the heart of an evolutionary computing application. It is responsible for determining which solutions within a population are better at solving the protein structure prediction problem in lattice model.

In this model, each amino acid is classified based on its hydrophobicity as either an H (hydrophobic or non-polar) or a P (hydrophilic or polar). The HP lattice model allows HP protein sequences to be configured as self-avoiding walks (SAW) on the lattice path favoring an energy free state due to HH interaction. The energy of a given

conformation is defined as the number of topological neighboring (TN) contacts between those Hs, which are not adjacent in the sequence.

Figure 2 shows an example for the 3D FCC HP lattice model.

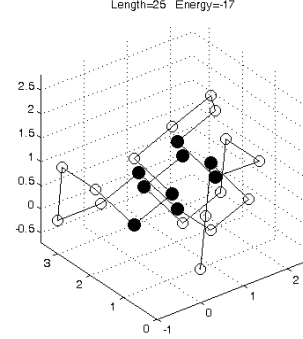


Figure 2. An optimal conformation for the sequence “PPHPPHHPPPPHPPPPHPPPHH” in a 3D FCC HP lattice model. The black filled dots denote the hydrophobic amino acid and the white open circles denote the polar amino acids. The H-H contacts (free energy) in the conformation, which are assigned the energy value of -1. The free energy is defined as a minimum value; the maximum number of H-H contact is given in the case of three-dimensional models. Figure 1 illustrates a protein structure with 17 H-H contacts (energy= -17).

2.3. HP Model Fitness Function

In the HP model, the free energy EHH for a protein structure can be calculated by using the fitness function below [9].

$$\epsilon_{ij} = \begin{cases} -1.0 & \text{the pair of H and H residues} \\ 0.0 & \text{others} \end{cases} \quad (1)$$

$$E_{HH} = \sum_{i,j} \Delta r_{ij} \epsilon_{ij}, \quad (2)$$

where the parameter

$$\Delta r_{ij} = \begin{cases} 1 & S_i \text{ and } S_j \text{ are adjacent but not connected amino acids} \\ 0 & \text{others} \end{cases} \quad (3)$$

Hence, the problem of the optimization of protein folding into the calculation of the minimal free energy of the protein folding conformation can be transformed. As a result, the following problem can be formally defined: given an HP sequence $s = s_1, s_2, \dots, s_n$, find an energy-minimizing conformation of s ; that is: find $c^* \in C(s)$ such that $E(c^*) = \min\{E(c) \mid c \in C\}$, where $C(s)$ is the set of all valid conformations for s [10].

2.4. HPC Model Fitness Function

In Equation (4) C_i and C_j represent the two cysteines in a disulfide bond. When two cysteines are correctly paired the free energy of the disulfide bond is set as -1. If they are not correctly paired, the free energy is set as 0. The objective function ECC as shown in Equation (5) is the sum of all the energy functions of the disulfide bonds.

$$\xi_{ij} = \begin{cases} -1.0 & \text{the pair of } C_i \text{ and } C_j \text{ residues} \\ 0.0 & \text{others} \end{cases} \quad (4)$$

$$E_{cc} = \sum_{i,j} \xi_{ij} \quad (5)$$

Figure 3 shows an example for the 2D FCC lattice HPC model. Furthermore, the predication approximation to the native physical protein structure (with short sequences) is significantly higher as illustrated in Figure 3.

Figure 3 a) the protein structure established experimentally b) predication of the known protein structure as in a) using 2D FCC lattice. In Figure 3 the conformation from HPC model after the addition of disulfide bonds between the 2nd and 12th residues and between the 3rd and 8th residues can be seen to be closer to the protein's native structure.

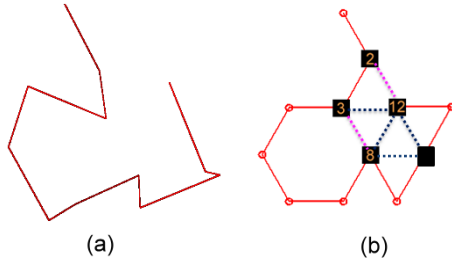


Figure 3. The protein structure established experimentally b) predication of the known protein structure as in a) using 2D FCC lattice HPC model.

2.5. Root-Mean-Square-Deviation (RMSD)

The RMSD is defined as follows: [11]

$$\text{RMSD} = \sqrt{\frac{\sum_{i=1}^{n-1} \sum_{j=i+1}^n (r_{ij}^{\text{model}} - r_{ij}^{\text{real}})^2}{\frac{n(n-1)}{2}}} \quad (6)$$

2.6. Polarity of Amino Acids

By analyzing large numbers of protein structures from protein databases, the Miyazawa-Jernigan (MJ) [11-12] matrix develops

this statistical potential by tabulating the interaction strength between any two types of amino acids (H and P) in proteins. MJ Matrix has thus gained considerable popularity in protein design and folding stimulations. Recently, Li et al. [13] rebuilt the MJ matrix and calculated the polarity of each amino acid as summarized in Table I.

TABLE I. POLARITY OF AMINO ACIDS

Hydrophobicity	Amino Acid
Hydrophobic (H)	A,D,E,G,H,K,N,P,Q,R,S,T
Hydrophilic or polar (P)	C,F,I,L,M,V,W,Y

3 Related Work

The optimization problem of protein folding and protein structure prediction by the HP model since it was proposed has attracted attention of many researchers who subsequently developed numerous solutions [14-22].

On the other hand if the disulfide bonds can be correctly predicted not only the searching capacity of protein conformations can be reduced but also the 3D tertiary structure of a protein can be predicted more accurately.

In the past other researchers have proposed many approaches on the correct and accurate prediction of disulfide bonds with high accuracy such as the 82% accuracy rate reported by Muskall et al., [23] and Fiser and Simon [24], the 90% accuracy rate reported by Chen et al., [25] and the 84% accuracy rate by Chen and Huang [26]. On top of that Ceroni et al., [27] developed DISULFIND server that offers prediction of disulfide bonds.

As a result though HPC model was used in our study the prediction of disulfide bonds it self is not included. The optimization problem of protein folding was studied with the assumption that all the disulfide bonds are known in the protein sequence. The emphasis of our study therefore is to improve the algorithms with the input of disulfide bonds to study particularly the impact of disulfide bonds on protein structure prediction.

4 Evolutionary Algorithms

The basic components of EA-based protein structure predication include crossover, mutation, local search, and fitness function. Figures 4 show the flowcharts of the proposed EA [37].

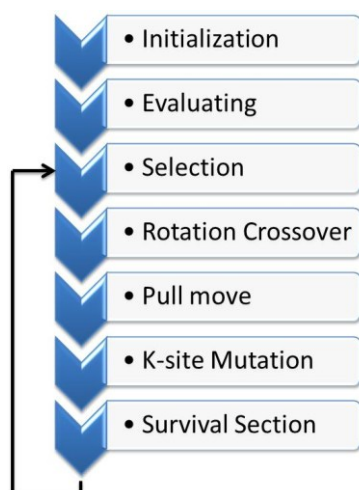


Figure 4. Flowchart of the proposed EA.

4.1. Initialization

An initial population is generated randomly and initializes at an $(n-1)$ -dimensional space within a fixed range. If the input amino acid sequence is of length n , then each individual in the population is a string of length $n - 1$ over the symbols = $\{1,2,3,4,5,6,7,8,9,10,11,12\}$, which represent $\{FL; FR; FU; FD; BL; BR; BU; BD; LU; LD; RU; RD\}$, and denote a valid conformation in the 3D FCC lattice.

4.2. Evaluating

The evaluating step evaluates each chromosome in a population. The goal of the EA method is to minimize the fitness value. The lower a fitness value, the better the fitness. The fitness function is used by equations (1-3) HP model and equations (1-5) HPC model.

4.3. Rotation Crossover

The idea of Rotation Crossover was first proposed by Unger and Moulton [14] and used on the 2D square lattice model. However, the 3D FCC lattice model is a very tight model composed of triangles and squares. Not all rotations on the matrix are effective as all the points must remain on the matrix after rotation.

Our recent study observes that every neighboring point of the centre point is in the same square and hexagon with any other neighboring ones. We therefore identify two different types of rotation: rotation along a square (type I) or a hexagon (type II). Each type can have different rotation angles. Furthermore, all the rotations can be represented by permutations of the labels of neighboring points. Our rotations are summarized as follows.

- Type I :

Rotations along a square with rotation angle 90° , 180° and 270° .

- Type II :

Rotations along a hexagon with rotation angle 120° and 240° .

During crossover, rotations are performed to increase success rate and to extend the exploration range of local search.

4.4. K-site Move Mutation

Single or double site move was also used frequently in previous local search methods [28-30]. After a complete and in-depth study on the methods of moves in local search, a complete k site move will be proposed. Together with a systemic Breadth-First-Search (BFS) and lower bounding, all likely coordinates of moves will be searched and serve as the basis for mutation and local search.

In FCC lattice model, every lattice has 12 neighboring points which move with self-avoiding walks. The K -site mutation in this study uses self-avoiding walks to move k number of sites. The complicity of SAW can be calculated as the formula [31]:

$$SAW_{fcc} = 1.26K^{0.16}(10.0364)^K \quad (7)$$

However the searching space increases with the increase of K . Therefore the lower bounding method used in our study can remove impossible searching space to reduce the complicity of searching.

When the K value is too low the searching space becomes smaller and the performance is limited. But if the K value is too high the searching space will be increased with the cost. Therefore K is set as 3 in our study.

4.5. Local Search of Pull Move

Pull Moves were first proposed by Lesh et al. [32] and used in the local search on the 2D square HP protein folding problem. Böckenbauer et al. [33] later applied pull move in 2D triangular and 3D FCC lattice models and proved that this method is reciprocal and complete. Pull Move as described in Böckenbauer et al. is identical with the method proposed by Lesh et al. [32]. Both methods pull the next site to its previous position. However, it is observed that in the FCC lattice

model pull move can be preserved without moving the next site to its previous position. In the 3D FCC lattice model there are four common neighboring points to be selected (between $+60^\circ$ and -60°). When operating pull move, if there is more than one position on the matrix to be chosen, a point can be chosen randomly for the pull moves.

4.6. Termination condition

The algorithm is run for a maximum of iterations. The best member of the population is then returned.

4.7. Parameter

All the parameter settings are listed in Table II.

TABLE II. PARAMETER SETTINGS

Operations/ Parameters	Setting
Representation	1-12 integer representative of {FL;FR;FU;FD;BL;BR;BU;BD;LU;LD;RU;RD}
Initialization	Random (SAW)
Crossover rate	0.8
Mutation rate	1/L
Parents selection	Tournament selection

5 Experimental Results

In this study, the benchmark is small proteins which contain the disulfide bond. The small proteins which contain the disulfide bond data are collected from the protein data bank (PDB). The benchmark of the study is listed on Table III

In order to prove the result of the study, the experiment is divided into three stages. Firstly, HP model is used to proceed with optimization experiment to give evidence that our approach can find the best global solution. Secondly, the fitness function of HPC model is applied in the stimulation experiment to search for the lowest free energy of disulfide bonds. Finally the stimulation results from the two models were compared with natural structure of the protein against RMSD.

TABLE III. THE BENCHMARKS FOR HP AND HPC MODEL

No.	PDB ID	Length	Disulfide
1	1CZ6	25	(4-20;10-16)
2	1EI0	38	(3-34;13-24)
3	1CRN	46	(3-40;4-32;16-26)
4	1EHS	48	(10-48;21-36)
5	1E8R	50	(6-37;16-31)

6	2ERS	66	(3-45;29-63)
---	------	----	--------------

5.1. Experimental results from HP model

Experimental results from HP model is shown in Table IV showing that the approach developed in this study can find the best solution using all the peptide sequences studied in this paper.

The Constraint-based Protein Structure Prediction (CPSP) approach is the best algorithm to find minimum-energy conformation on the HP lattice model [34-36]. However when the experimental peptide sequence cannot converge, no close answer can be obtained. For example PDB ID: 4BP2, 2SOD, 1TNF, 9WGA, 1BBP, 1RBP and 4SGB et al., In addition this approach is not suitable for any 20 amino acid pairwise interactions energy function. Consequently Evolutionary Algorithms (EA) so far has been the most widely used approach.

TABLE IV. EXPERIMENTAL RESULTS FROM HP MODEL. THE FORMAT OF COLUMN ENTRIES IS 'AVERAGE / MINIMUM'. NUMBERS IN BOLD INDICATE THE LOWEST ENERGY. IN ALL CASES, NATIVE ENERGY IS COMPUTED USING CPSP-TOOLS HPSTRUCTURE, DESCRIBED IN [34-36]. THE SEQUENCE 1 TO 2 IF THE POPULATION SIZE WAS SET AS 10 AND THE GENERATION NUMBER AS 30. FOR SEQUENCE 3 TO 6 THE BEST SETTING IS 50 FOR POPULATION SIZE AND 150 FOR GENERATION NUMBER.

No.	PDB ID	Length	E*	Our Method
1	1CZ6	25	-21	-19.53/-21
2	1EI0	38	-29	-26.37/-29
3	1CRN	46	-45	-43.27/-45
4	1EHS	48	-35	-33.00/-35
5	1E8R	50	-46	-43.80/-46
6	2ERS	66	-68	-63.93/-68

5.2. Experimental results from HPC model

Table V summarizes the stimulation results obtained by fitness function using the HPC model. Due to the insertion of disulfide bonds the results from the HPC model were different from the HPC model and the free energy would be different too. In addition a best global solution cannot be promised using the proposed approach in HPC model as in this study but the closest best solution would be obtained.

TABLE V. EXPERIMENTAL RESULTS FROM HPC MODEL

No.	PDB ID	Length	E_{HH}	E_{CC}
1	1CZ6	25	-18.23/-21	-2
2	1EI0	38	-23.07/-29	-2
3	1CRN	46	-34.47/-40	-3
4	1EHS	48	-25.63/-33	-2
5	1E8R	50	-36.87/-43	-2

6	2ERS	66	-47.17/-58	-2
---	------	----	------------	----

5.3. RMSD Comparison

Table VI shows the RMSD in the HP and HPC models. It can be found from Table VI that the HPC model yielded better stimulations in all the protein sequences with the introduction of disulfide bonds than the HP model. This suggests that the presence of disulfide bonds can improve the computational prediction of 3D protein structure.

TABLE VI. COMPARISON OF RMSD IN THE HP AND HPC MODELS, MONOMER DISTANCE OF 3.8 ÅNGSTROM.

No.	PDB ID	Length	HP model	HPC model
1	1CZ6	25	13.25	11.08
2	1EI0	38	10.09	9.45
3	1CRN	46	11.38	9.60
4	1EHS	48	13.29	9.72
5	1E8R	50	9.62	9.30
6	2ERS	66	14.99	13.49

5.4. Visual comparison

Finally the stimulation results from HP and HPC models were compared visually as shown in Figure 5. In Figure 5 the conformation from HPC model after the addition of disulfide bonds can be seen to be closer to the protein's native structure than the prediction using the HP model (Figure 5 b & c). This shows that the addition of disulfide bonds into the algorithms can indeed improve the prediction performance.

However due to the time limit the sample numbers used in this experiment were small. As a result more experiments need to be carried out to prove the correlation between the addition of disulfide bonds and the corresponding prediction performance.

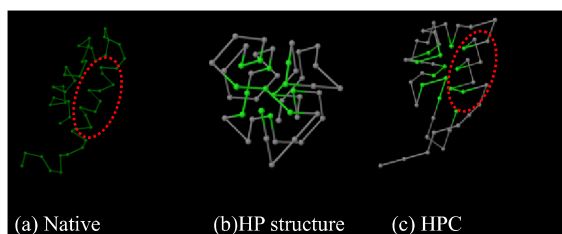


Figure 5. Visual comparison for PDB id: 1EHS.

(a) is a real protein structure; (b) is HP model minimum-energy conformations and (c) is HPC model simulation result. In (b) and (c), green balls indicate hydrophobic amino acids while the gray balls indicate the hydrophilic amino acids.

6 Conclusions

This paper study the protein structure prediction issue, and propose an evolutionary algorithm to

yield improve similarity between the predicted result and the real protein structure for all sequences in the benchmarks. Specifically, a new hydrophobic-polar-cysteine (HPC) model is integrated with the fitness function of the evolutionary algorithm. The proposed lattice rotation crossover and the K-site move mutation can help the algorithm find the global optima. Experiment results show that the proposed approach is able to find minimum-energy conformations.

In addition, the study differs from HP model by adding the Cysteine into the model. Because the disulfide bond is a kind of covalent bond so the force between the bonds is stronger than the normal hydrogen bonds and affects the protein structure much. The experiment indicates we can get much more similar structure on forecasting the protein structure after adding the Cysteine residues into the protein sequence to form disulfide bonds. In the future we will perform more experiments for protein sequence with disulfide bonds such as RS126 and CB513 dataset to gain more valuable insight in the research of protein structure prediction. We hope to increase the similarity on forecasting the protein structure and prove the feasibility of HPC model.

References

- [1] K. A. Dill, "Theory for the folding and stability of globular proteins," *Biochemistry*, vol. 24, 1985, pp. 1501–1509.
- [2] K.F. Lau, and K.A. Dill, "A Lattice Statistical Mechanics Model of the Conformational and Sequence Spaces of Proteins," *Macromolecules*, vol. 22, 1989, pp. 3986–3997.
- [3] P.Crescenzi et al., "On the complexity of protein folding," *J. Comp. Biol.*, vol.5, 1998, pp. 409–422.
- [4] B. Berger and T. Leight, "Protein folding in the hydrophobic-hydrophilic (HP) model is NP-complete," *J. Comp. Biol.*, vol.5, 1998, pp. 27–40.
- [5] R. Jaenicke, "Protein stability and molecular adaptation to extreme conditions," *European Journal of Biochemistry*, 1991, pp.715–728
- [6] A. Hadj Khodabakhshi, J. Manuch, A. Rafiye, A. Gupta, "Structure approximating design of stable proteins in 2D HP model fortified by cysteine monomers," *In proc. of The Sixth Asia Pacific Bioinformatics Conference (APBC08)*, Kyoto, 2008
- [7] A. H. Khodabakhshi et al., "Inverse protein folding in 3D hexagonal prism lattice under HPC model," *J Comput. Biol.* 16(6), 2009, pp.769–802.
- [8] G. Raghunathan, R. L. Jernigan, "Ideal architecture of residue packing and its observation in protein structures," *Protein Science*, 1997, pp. 2072 – 2083.
- [9] C.Huang,X. Yang, and Z.He,"Protein folding simulations of 2D HP model by the genetic algorithm based on optimal secondary structures," *Computational Biology and Chemistry*, 2010, pp. 137–142.
- [10] A. Shmygelska and H. H. Hoos, "An ant colony optimisation algorithm for the 2D and 3D

- hydrophobic polar protein folding problem,”** *BMC Bioinformatics*, pp. 30, 2005
- [11] S. Miyazawa, and R. L. Jernigan, “**Estimation of effective interresidue contact energies from protein crystal structures: quasi-chemical approximation,”** *Macromolecules*, 1985, pp. 534–552.
- [12] S. Miyazawa, and R. L. Jernigan, “**Residue-residue potentials with a favorable contact pair term and an unfavorable high packing density term, for simulation and threading,”** *J. Mol. Biol.*, 1996, pp. 623–644.
- [13] H. Li, C. Tang, and N. S. Wingreen, “**Nature of Driving Force for Protein Folding: A Result From Analyzing the Statistical Potential,”** *Phys. Rev. Lett.*, 1997, pp. 765–768.
- [14] R. Unger, and J. Moulton, “**Genetic algorithms for protein folding simulations,”** *Journal of Molecular Biology*, 1993, pp. 75–81.
- [15] T. Jiang et al., “**Protein folding simulations for the hydrophobic-hydrophilic model by combining tabu search with genetic algorithms,”** *Journal of Chemical Physics*, 2003, pp. 4592–4596.
- [16] M. T. Hoque, M. Chetty, and A. Sattar, “**Protein Folding Prediction in 3D FCC HP Lattice Model Using Genetic Algorithm,”** in *Bioinformatics special session, IEEE Congress on Evolutionary Computation (CEC)*, Singapore, 2007, pp. 4138–4145.
- [17] M. T. Hoque, M. Chetty, A. Sattar, “**Extended HP Model for Protein Structure Prediction,”** *Journal of Computational Biology*, Vol. 16(1), 2009, pp. 85–103
- [18] S.-C. Su, C.-J. Lin and C.-K. Ting, “**An effective hybrid of hill climbing and genetic algorithm for 2D triangular protein structure prediction,”** *Proteome Science*, 2011, vol. 9, S19, doi:10.1186/1477-5956-9-S1-S19
- [19] C.-J. Lin, and S.-C. Su, “**Protein 3D HP Model Folding Simulation Using a Hybrid of Genetic Algorithm and Particle Swarm Optimization,”** *International Journal of Fuzzy Systems*, Vol. 13, No. 2, 2011, pp. 140–147.
- [20] M. T. Hoque et al., “**Twin-Removal in Genetic Algorithms for Protein Structure Prediction using Low Resolution Model,”** *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2011, pp. 234 – 245.
- [21] J.-J. Tsay, and S.-C. Su, “**A Memetic Algorithm for Protein Structure Prediction Based on 2D Triangular Lattice Model,”** *accepted for publication on Bioinformatics Conference*. 2012.
- [22] J.-J. Tsay, S.-C. Su, “**Ab initio protein structure prediction based on memetic algorithm and 3D FCC lattice model,”** *Bioinformatics and Biomedicine Workshops (BIBMW)*, 2011 IEEE International Conference, pp. 315–318
- [23] S. M. Muskal et al., “**Prediction of the disulfide-bonding state of cysteine in proteins,”** *Protein Eng.*, 3, 1990. pp. 667–672.
- [24] A. Fiser, and I. Simon, “**Predicting the oxidation state of cysteines by multiple sequence alignment,”** *Bioinformatics*, 16, 2000, pp. 251–256.
- [25] A. Ceroni et al., “**DISULFIND: a Disulfide Bonding State and Cysteine Connectivity Prediction Server,”** *Nucleic Acids Research*, 34(Web Server issue), 2006, W177–W181.
- [26] Y.-C. Chen, and J.-K. Hwang, “**Prediction of Disulfide Connectivity from Protein Sequences,”** *Proteins: Structure, Function, and Bioinformatics*, 2005, pp. 507–512.
- [27] C. Thachuk, A. Shmygelska and H. H. Hoos, “**A replica exchange Monte Carlo algorithm for protein folding in the HP model,”** *BMC Bioinformatics*, 2007, 8:342
- [28] P. H. Verdier, W. H. Stockmayer, “**Monte Carlo Calculations on the Dynamics of Polymers in Dilute Solution,”** *The Journal of Chemical Physics*, 1962, 36:227–235.
- [29] K. A. Dill, S. Bromberg, K. Yue, K. M. Fiebig, D. P. Yee, P. D. Thomas, and H. S. Chan, “**Principles of Protein Folding - A Perspective From Simple Exact Models,”** *Protein Science*, 1995, pp. 561–602.
- [30] M. T. Gurler, C. C. Crabb, D. M. Dahlin, J. Kovac, “**Effect of bead movement rules on the relaxation of cubic lattice models of polymer chains,”** *Macromolecules*, 1983, 16(3):398–403.
- [31] P. Schuster, P. F. Stadler, “**Discrete Models of Biopolymers,”** In *Handbook of Computational Chemistry* Edited by: Crabbe MJC, Drew M, Konopka A. Marcel Dekker, New York; 2001.
- [32] N. Lesh, M. Mitzenmacher, and S. Whitesides, “**A complete and effective move set for simplified protein folding,”** In *RECOMB '03: Proceedings of the seventh annual international conference on Research in computational molecular biology* New York, NY, 2003.
- [33] H.-J. Böckenhauer et al., “**A Local Move Set for Protein Folding in Triangular Lattice Models,”** *Algorithms in Bioinformatics*, LNCS, 2008, pp. 369–381.
- [34] R. Backofen and S. Will, “**A constraint-based approach to fast and exact structure prediction in three-dimensional protein models,”** *Journal of Constraints*, 2006, pp. 5–30.
- [35] M. Mann, S. Will and R. Backofen, “**CPSP-tools - Exact and Complete Algorithms for High-throughput 3D Lattice Protein Studies,”** *BMC Bioinformatics*, 9, 2008, 230.
- [36] M. Mann et al., “**CPSP-web-tools: a server for 3D lattice protein studies,”** *Bioinformatics* 2009, pp. 676–677.
- [37] S.-C. Su, J.-J. Tsay, “**Rotation Crossover and K-Site Move Mutation for Evolutionary Protein Folding in 3D FCC HP Model,”** *Bioinformatics and Biomedicine*, 2011, IEEE International Conference.